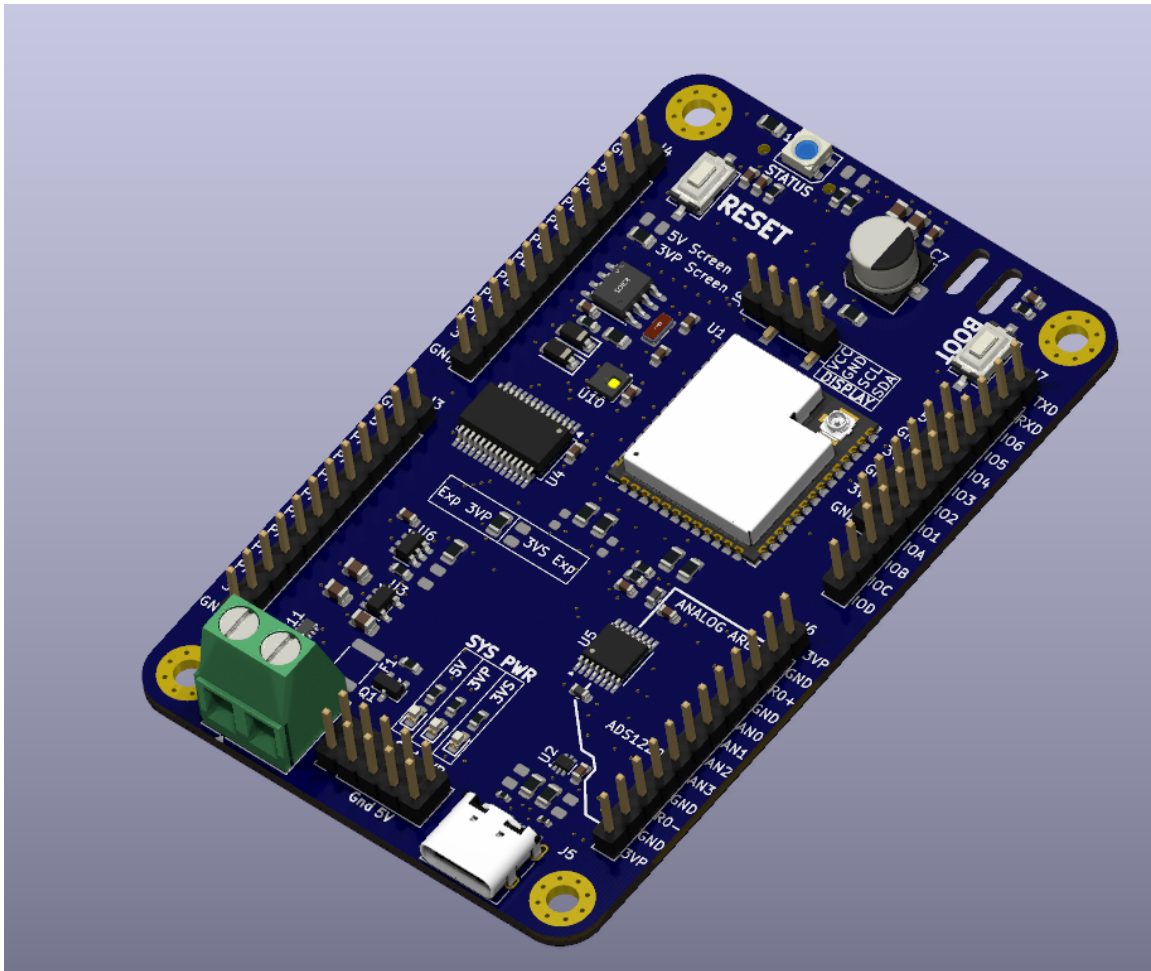


NWU Engineering ESP32-S3 Dev Board

v1.1 | Designed by Prof H Marais | 2026

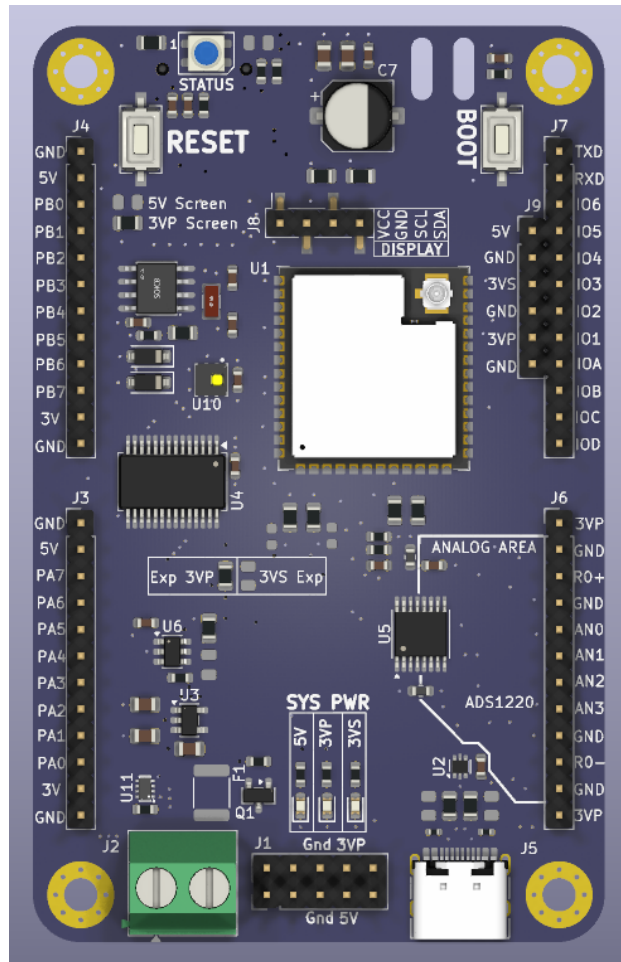


The board's ready. Here's what's on it and what it's for.

This is a general-purpose ESP32-S3 development board built for the course — capable of driving real motor/inverter control work out of the box, while also functioning as a complete, self-contained datalogger: onboard SD card, real-time clock, and a precision sensor front end, all on one board with plenty of room to expand.

Core

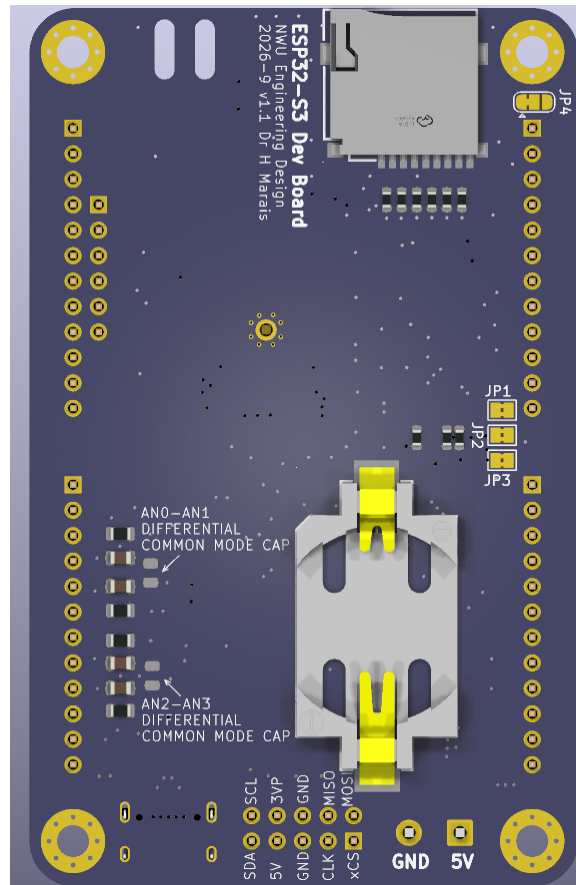
- **ESP32-S3-WROOM-1U** module — 16MB flash, 8MB PSRAM, dual-core, Wi-Fi + BLE, native USB
- **External antenna** via a pigtail — the board doesn't rely on Wi-Fi range from an onboard antenna, useful once it's mounted inside an enclosure
- **USB-C** for power and programming, plus a **screw-terminal 5V input** for external power — both properly protected against reverse polarity and against backfeeding one supply into the other
- **RESET** and **BOOT** buttons, a status LED, and three small power-indicator LEDs so you can tell at a glance what's actually powered



Top view — header and jumper labels

Built-in Peripherals

- **MCP23017 I/O expander** — 16 extra digital I/O pins over I2C
- **ADS1220 precision ADC** — 24-bit, 4 channels, configurable gain, differential or single-ended — a proper analog front end for sensor work, not just the chip's own noisy internal ADC
- **AHT20** temperature/humidity sensor
- **PCF8563 real-time clock**, with a coin-cell backup — the board keeps accurate time across power cycles, no network sync required
- **microSD card slot** — for onboard datalogging
- **I2C display connector**, with a jumper to select 5V or 3.3V, for hanging a small OLED/TFT off the board directly



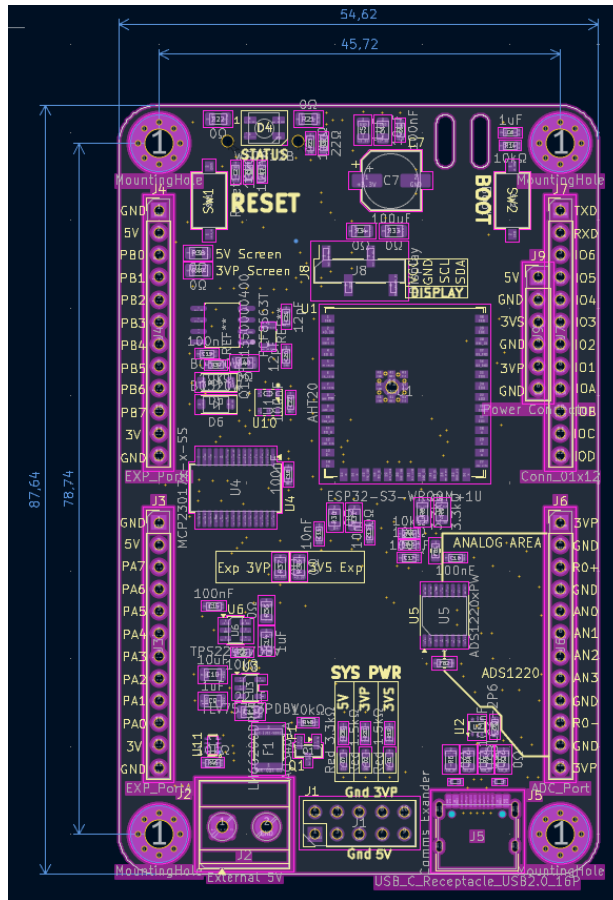
Back view — SD card slot and RTC coin-cell holder

Three Expansion Headers, Each Built for a Different Job

- **Digital I/O header** — runs through the MCP23017, not the ESP32 directly. Good for general switching, buttons, relays — and it protects the microcontroller itself if something gets miswired.
- **ESP IO header** — direct microcontroller GPIO, including 6 PWM channels wired for real motor/inverter control (deadtime-capable, with a dedicated fault/E-stop input), plus general-purpose pins for whatever else you need close to the chip.
- **Analog header** — the ADC's 4 channels plus reference pins, for precision measurement work: thermocouples, RTDs, load cells, and similar.

One More Thing Worth Knowing

Every header and mounting hole is deliberately aligned to the standard 0.1" (2.54mm) prototyping grid — so the board plugs straight into ordinary perfboard or stripboard for anything you want to build on top of it.



Board outline and mounting-hole spacing

Questions or issues once you've got a board in hand — usual channels.